

P1091R2

Extending structured bindings to be more like variable declarations

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This version:

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Audience:

CWG

Project:

ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Abstract

Making structured bindings more like first-class-citizen.

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§ 1. This revision

This revision contains only wording for the changes approved by EWG:

- By value capture of structured bindings
- static and thread_local for structured binding declarations

Read a previous revision to know more.

§ 2. Proposed wording

Change [expr.prim.lambda.capture]p8 (7.5.5.2) as follows:

If a *lambda-expression* explicitly captures an entity that is not odr-usable ~~or captures a structured binding (explicitly or implicitly)~~, the program is ill-formed.

Change [expr.prim.lambda.capture]p12 (7.5.5.2) as follows:

A bit-field a structured binding, or a member of an anonymous union shall not be captured by reference.

Change [dcl.struct.bind]p1 (9.5) as follows:

If the *assignment-expression* in the *initializer* has array type A and no *ref-qualifier* is present, ~~e has type cv A~~ is defined by

*attribute-specifier-seq*_{opt} *decl-specifier-seq* cv A *e*;

where the *decl-specifier-seq* does not include the *type-specifier* auto and each element is copy-initialized or direct-initialized from the corresponding element of the *assignment-expression* as specified by the form of the *initializer*.

Add a sentence to [dcl.struct.bind]p1 (9.5) as follows:

If either the *storage-class-specifier* static or thread_local appears in the *decl-specifier-seq* of the structured binding declaration, then the declaration shall not be defined at namespace scope.

Change [dcl.dcl]p8 (10) as follows:

The *decl-specifier-seq* shall contain only the *storage-class-specifiers* static and thread_local, the *type-specifier* auto and *cv-qualifiers*.

Change [dcl.stc]p3 (10.1.1) as follows:

It shall be applied only to the names of variables of namespace or block scope, to structured binding declarations ([dcl.struct.bind]), and to the names of static data members.

Change [dcl.stc]p4 (10.1.1) as follows:

The **static** specifier can be applied only to names of variables and functions, to structured binding declarations ([dcl.struct.bind]), and to anonymous unions.